

Game Bear: Upgrading gaming experiences worldwide with data analytics and a powerful global network

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ABOUT GAME BEAR



To enhance the quality of its cross-border gaming experiences, Game Bear uses the global network, translating data analytics capabilities of Google Cloud to improve gaming experiences.

About Game Bear

Founded in 2015, Game Bear is a mobile game developer and distributor based in Hong Kong specialized in open-world multiplayer sci-fi games. Currently offering three games to more than 10 millions users around the world, Game Bear is dedicated to building game worlds where gamers can freely design their own gameplays and interact with fellow players across borders.

Industries: Gaming

realize more efficient
effective decision

Location: Hong Kong

Compute Engine

Google Cloud results

- Lowers network latency by up to 10X worldwide through Cloud Load Balancing
- Supports efficient and effective data-driven decision making with Looker
- Helps realize more objective A/B testing while reducing time and resources required

Shorter
analytical
generation
from cloud
to one



About WebEye

WebEye integrates global resources and uses the best technology to build a global digital growth ecosystem around 'content,' 'traffic,' and 'data,' helping all enterprises to connect to the world market.

WebEye provides a range of customized services such as cloud consulting, cloud migration and cloud management to accelerate the digital transformation of enterprises. In 2021, WebEye achieved Google Cloud Managed Services Provider (MSP) status and was the winner of the 2021 Google Cloud Breakthrough Partner for Asia Pacific, based on its extensive technical capabilities in Google Cloud solutions and products.

connect to the world market.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

services such as cloud consulting, cloud

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Google Cloud Managed Services Provider

with
Firebase
• Enables 20-
30% higher
ad
conversion
rate with ad
performance
predictions
provided by
Vertex AI

Compute Engine

(<https://cloud.google.com/compute/>)

for Asia Pacific, based on its extensive
Firebase (<https://firebase.google.com/>)
technical expertise in Google Cloud

Looker (<https://cloud.google.com/looker>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Cloud Translation API

(<https://cloud.google.com/translate/>)

Open-world games are one of the most popular genres of online video games globally, because gamers can enjoy a high level of freedom to interact with other players, and explore different gameplays or even create their own ones. To fully immerse themselves in game worlds, gamers not only look for fascinating visual effects, but also support for smooth network connectivity and communication among fellow players.

With its mission to create cross-border and diverse game worlds, Game Bear has been improving the gaming experiences that it offers to gamers. Founded in 2015, the Hong Kong-based mobile game developer and distributor specializes in open-world multiplayer sci-fi games, and currently offers three games to more than 10 million users worldwide. Its flagship game, [Nova Empire: Space Commander](https://play.google.com/store/apps/details?id=com.gamebeartech.nova)

(<https://play.google.com/store/apps/details?id=com.gamebeartech.nova>)

is known for its beautiful graphics of space scenes and intriguing strategy system.

"We want to create game worlds together with our users, which is why we're dedicated to providing the greatest freedom possible to gamers to design their own gameplays and interact with fellow players. Eliminating geographical and language barriers is an important part of that effort," explains Yuan Chen, CEO of Game Bear.

"To improve the quality of our games and the way we implemented our data-driven decision-making process, we needed a better-connected global network, a translation tool that delivers highly accurate results, and data analytics

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—Yuan Chen, CEO, Game
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When Game Bear released its first game, it deployed the game on a cloud platform hosted in Hong Kong. To offer global connectivity to gamers around the world, Game Bear needed to set up and manage network nodes in different continents manually, which required more workforce than it could afford and resulted in high and unstable latency. At the same time, the translation and data analytics tools that Game Bear was using couldn't meet its demand for accurate translations and efficient data analyses. To enhance the gaming experiences that it offered and its operations efficiency, the game developer migrated its game to Google Cloud (<https://cloud.google.com/>) in 2016 for its extensive global network and high-performance translation and data analytics tools.

"To improve the quality of our games and the way we implemented our data-driven decision-making

process, we needed a better-connected global network, a translation tool that delivers highly accurate results, and data analytics software that supports near real-time analytics," notes Chen.

"Google Cloud meets our needs in all these respects."

Building cross-border game worlds with global network and Translation AI

Following the advice provided by Google Cloud partner WebEye

(<https://www.webeye.com/english/home.html>) to use the private network of Google Cloud to transfer its user data across continents, Game Bear successfully migrated its game from the cloud platform that it previously used to Google Cloud in three hours, instead of one day if it used the public network.

From then on, the game developer has been launching its new games on Google Cloud. Now, Game Bear uses the virtual machines (VMs) on Compute Engine (<https://cloud.google.com/compute>) as the servers of its games, and Cloud SQL (<https://cloud.google.com/sql>) as its backup database.

To improve the network quality of its games worldwide, Game Bear adopts Cloud Load Balancing (<https://cloud.google.com/load-balancing>), which according to Chen delivers the lowest latency for all the internet protocols that the game developer uses compared to other similar offerings. Before the migration, Game Bear relied on an external team of 10 engineers to manage the network nodes around the world that it deployed, but the network quality

was still far from ideal, with average latency of 400-500 microseconds, a daily disconnection rate of 5% and frequent data loss during high traffic. Backed by the global network of Google Cloud, Cloud Load Balancing has helped Game Bear significantly enhance the connectivity of its games through simple settings. Not only does the game developer no longer need to deploy and manage global network nodes by itself, the latency of its games has been reduced to as low as 40 microseconds, with 0.5% daily disconnection rate and no data loss.

"Cloud Load Balancing is now the backbone of our network service. With the support of the extensive global network of Google Cloud, we're able to greatly improve the gaming experiences of our game for users around the world by offering smooth and reliable connectivity," says Chen.

On top of that, the highly accurate translations among more than 20 languages generated by [Translation AI](https://cloud.google.com/translate) (<https://cloud.google.com/translate>) have also become a fundamental pillar of Game Bear's cross-border game worlds, where gamers speaking different languages can easily communicate with each other. Based on the comparison test that the Game Bear team conducted, Translation AI delivers 50% more accurate and fluent translations, particularly for some languages that many Game Bear's users speak but are less widely used like Polish, Russian and Ukrainian.

Realizing powerful data-driven decisions with BigQuery and Looker

Since the release of its first game, Game Bear has been analyzing its user and business data to improve user experience and its operations. At the beginning, it used a third-party data analytics tool and built its own data warehouse. However, the data analytics tool was not fully compatible with Game Bear's data warehouse. It therefore required a long time to transfer data and generate analytics reports. To realize near real-time analytics for higher operations efficiency, Game Bear now uses Firestore (<https://firebase.google.com/>) to gather ad and user data, which are then automatically synchronized in BigQuery (<https://cloud.google.com/bigquery>) for data analysis. It used to take Game Bear's data team one week to produce a performance report for an ad, due to the different settings that had to be made manually. With Firestore and BigQuery, the same report can be generated in an hour.

Furthermore, Game Bear leverages Looker (<https://cloud.google.com/looker>) to visualize data and provide data insights during internal meetings. Before using Looker, Game Bear's data team had to be present at decision-making meetings to pull out data upon request. Since Looker offers intuitive interfaces, Game Bear's employees can easily retrieve any data on their own and run further data analyses on the spot. This way, the Game Bear team is able to make more effective business decisions in a short period of time by quickly extracting data insights.

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For example, Game Bear once found that an ad display had resulted in a decrease in user retention rate. At first, the team considered suspending ad displays for all users, but through further data analyses on Looker, it found out that the ad display only affected user retention in certain countries. By only stopping showing ads to users in these countries, Game Bear successfully resolved the user retention issue without sacrificing unnecessary revenue loss.

"The benefits that we've received from using BigQuery and Looker are inestimable," says Chen.

"Not only can we realize near real-time data analytics with minimal workforce, the insights that we pull from data have also helped us notice operations issues at an early stage and take effective actions in time."

Supporting cost-efficient A/B testing and ad performance tracking with Firebase

Besides data collection, Game Bear also employs Firebase for A/B testing to improve the features of its games. Previously, Game Bear's front-end and back-end engineers, as well as its operations team all had to be involved to conduct an A/B test, which required minimum one week to generate test results. Since the testing environments were set up manually, the objectivity of the tests was sometimes questioned. With Firebase, Game Bear only needs one engineer to configure the parameters needed, and the outcome can be generated in two days. At the same time, the fact that the testing

environments are built the same way in Firebase guarantees a high level of objectiveness.

Firebase has also helped Game Bear improve the return on investment of its ads through the [Dynamic Links](https://firebase.google.com/docs/dynamic-links) (<https://firebase.google.com/docs/dynamic-links>) feature, which enables the game developer to easily retrieve data from third-party systems and track the performance of its ads in different channels. This way, Game Bear can identify the ad channels that bring in more valuable users and adjust its ad resources accordingly.

Leveraging AI to optimize ad conversion and broaden game features

Currently, Game Bear uses [Vertex AI](https://cloud.google.com/vertex-ai) (<https://cloud.google.com/vertex-ai>) to predict the long-term conversion rate of its ads based on the performance figures of the first two to three days after release. Using the predictions generated by artificial intelligence (AI) as reference, the game developer is able to only distribute ads that deliver higher conversion and increase its overall ad conversion rate by 20 to 30%. Moving forward, it plans to expand its use of the AI capabilities of Google Cloud to improve its game features and develop AI-based games, such as adding AI-powered non-player characters.

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